Magneto-Optics Using Faraday Effect
Experiment 04

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1 Aim

- To estimate Verdet constant and check how rotation angle of the polarisation plane is changed by the magnetic flux.
- To verify Malus' law for the relation between .

2 Apparatus

The required apparatus for performing the experiment are:

1. Laser source
2. Electromagnet
3. Polarisers
4. Photodiode detector

3 Theory

The experiment is based on the opto-magnetic property, demonstrating Faraday effect which states that isotropic media become optically active when a magnetic field is applied in the propagation direction of the light. A linearly polarised light propagating along the x direction can be expressed as a superposition of left and right circularly polarised components.

$$\mathbf{E}(x,t) = \mathbf{E}_{\mathrm{r}}(x,t) + \mathbf{E}_{\mathrm{l}}(x,t) \tag{1}$$

where $\mathbf{E}_{\mathrm{r}}$ and $\mathbf{E}_{\mathrm{l}}$ denote the left and right circular components. Written explicitly we have,

$$\begin{aligned} \mathbf{E}_{\mathrm{r}}(x,t) &= \left(0, \frac{A}{2}\cos\left[\omega(t - n_{\mathrm{r}}x/c)\right], \frac{A}{2}\sin\left[\omega(t - n_{\mathrm{r}}x/c)\right]\right)^{\top} \\ \mathbf{E}_{\mathrm{l}}(x,t) &= \left(0, \frac{A}{2}\cos\left[\omega(t - n_{\mathrm{r}}x/c)\right], -\frac{A}{2}\sin\left[\omega(t - n_{\mathrm{r}}x/c)\right]\right)^{\top} \end{aligned} \tag{2}$$

where n_{r} and n_{l} are the different refractive indices for the RCP and LCP components. Let us choose $x = D$, and using the above relation we will obtain:

$$\psi = \tan^{-1}\left(\frac{E_{\mathrm{z}}}{E_{\mathrm{y}}}\right) = \frac{\omega}{2c}(n_{\mathrm{l}} - n_{\mathrm{r}})D \implies \psi = (n_{\mathrm{l}} - n_{\mathrm{r}})\frac{\pi D}{\lambda} \tag{3}$$

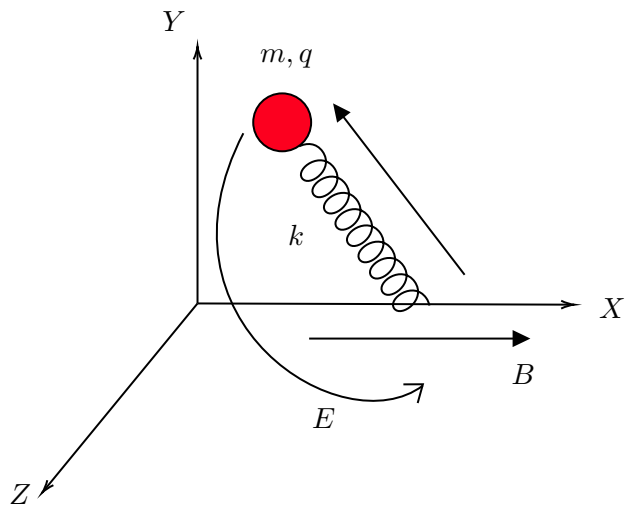
This shows that the difference in the refractive indices can rotate the linearly polarised light. Now, we show that the introduction of a magnetic field will change the refractive index for the two polarisation components using Becquerel's theory.

Consider a material containing particles of mass m and charge q, embedded in an opposite charge distribution. The system can be modelled as the masses being contrained by springs and vibrating about fixed sites. Considering magnetic field along the x axis and a plane RCP rotating clockwise with respect to the field, the equation of motion can be written as,

$$-m\omega^2 r_{\mathrm{rcp}} = -kr + Eq + Bq\omega r \implies r = \frac{Eq/m}{(\omega_0^2 - \omega^2 - Bq\omega)} \tag{4}$$

where $\omega_0^2 = \left(\frac{k}{m}\right)^{1/2}$ to be the natural frequency. Similarly, for the left circularly polarised light we can find,

$$r_{\text{lcp}} = \frac{Eq/m}{(\omega_0^2 - \omega^2 + Bq\omega)} \tag{5}$$



The displacement of the charge creates a dipole moment qr. Suppose there are N dipoles per volume in the system. Then the polarisation per volume is given by $P = Nqr$. Considering linear polarisation, $P = \varepsilon_0 \chi_{\text{e}} E$ where ε_0 is the permittivity of the free space and χ_{e} is the electric susceptibility. Now, the relative permittivity is, $\varepsilon_{\text{r}} = \chi_{\text{e}} + 1 = 1 + \frac{P}{\chi_0 E}$. From the value of the relative permittivity $n = \sqrt{\varepsilon_{\text{r}} \mu_{\text{r}}}$. Taking $\mu_{\text{r}} = 1$, we get:

$$\begin{aligned} n_{\text{r}}(\omega) &= \left[1 + \frac{Nq^2}{m\left(\omega_0^2 - \omega^2 - \frac{Bq\omega}{m}\right)}\right]^{1/2} \\ n_{\text{l}}(\omega) &= \left[1 + \frac{Nq^2}{m\left(\omega_0^2 - \omega^2 + \frac{Bq\omega}{m}\right)}\right]^{1/2} \end{aligned} \tag{6}$$

Considering the approximation, $\omega^2 + \frac{Bq\omega}{m} = (\omega + \Delta\omega_{\text{r}})^2$ and $\omega^2 - \frac{Bq\omega}{m} = (\omega + \Delta\omega_{\text{l}})^2$ where $\Delta\omega_{\text{r}} = -\Delta\omega_{\text{l}} = \frac{Bq}{2m}$ is considered to be small, we get:

$$n_{\text{l}} - n_{\text{r}} = \frac{\mathrm{d}n}{\mathrm{d}\omega} \cdot (\Delta\omega_{\text{l}} - \Delta\omega_{\text{r}}) = \frac{\mathrm{d}n}{\mathrm{d}\lambda} \cdot \frac{\mathrm{d}\lambda}{\mathrm{d}\omega} \cdot \left(-\frac{Bq}{m}\right) = \frac{\mathrm{d}n}{\mathrm{d}\lambda} \cdot \left(\frac{\lambda^2}{2\pi c}\right) \cdot \left(\frac{Bq}{m}\right) \tag{7}$$

This implies that the optical rotation can be written, $\Psi = VDB$ where V is the Verdet constant,

$$V = \frac{\mathrm{d}n}{\mathrm{d}\lambda} \cdot \frac{\lambda}{2c} \cdot \frac{q}{m} \tag{8}$$

The figure below shows the setup which we used for our experiments, along with all the optical elements.

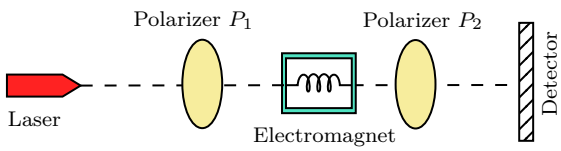


Figure 1: Schematic of the Experimental Setup

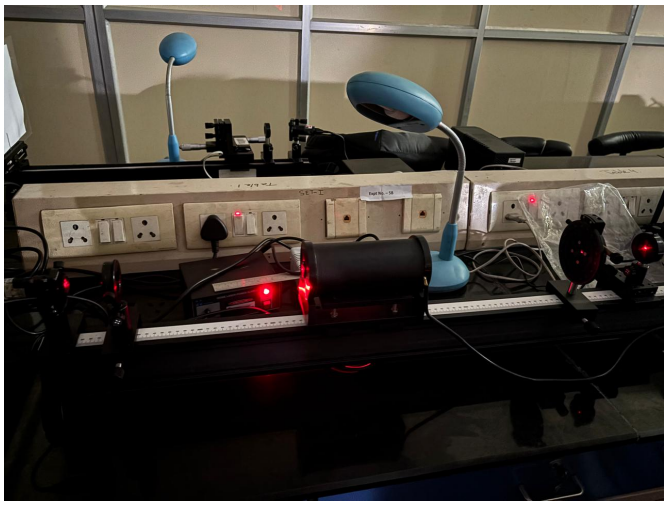


Figure 2: Actual experimental setup

We also verify Malu's law, which states $\mathcal{I} = \mathcal{I}_0 \cos^2(\Psi)$ and Ψ is proportional to the current I. For small values of optical rotation, we have a quadratic relation between intensity and current, $\mathcal{I} \sim I^2$.

4 Data and Calculation

The experiment consists of two parts, which we performed.

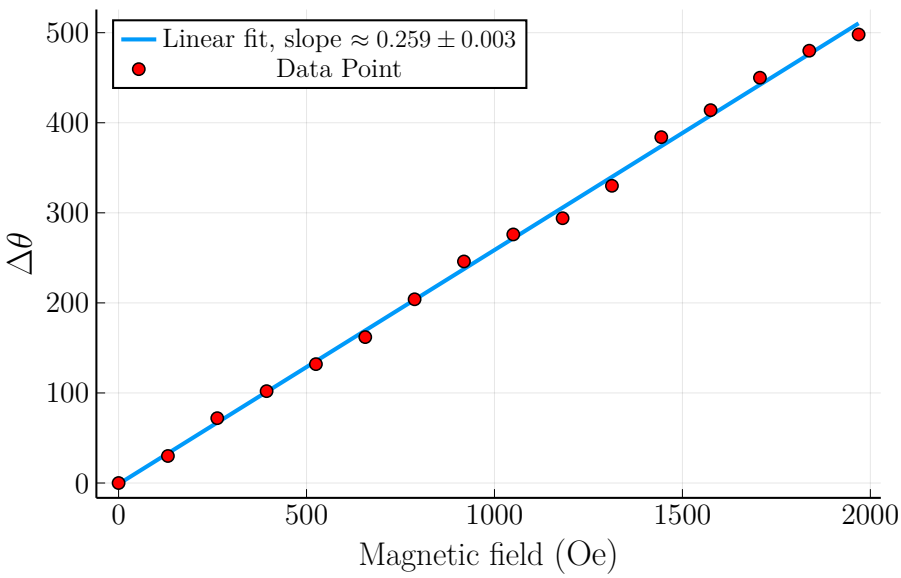
4.1 Estimation of Verdet's constant

In the first part, we put the two polarisers, one in between the electromagnet and laser source and the other between the screen and the electromagnet. We rotated the second polariser axis and measured the intensity obtained. For our experiment, the number of turns per unit length was specified as $N = 167.2/$ cm and length of the glass rod is 15 cm. The wavelength of the laser is $\lambda = 650$ nm

I (A)	$H = \pi N I$ Oe	θ (°)	$\Delta\theta$ (°)	$\Delta\theta$ (min)	V
0.00	0.00	296.90	0.00	0.00	–
0.25	131.25	296.40	0.50	30.00	0.02286
0.50	262.50	295.70	1.20	72.00	0.02743
0.75	393.76	295.20	1.70	102.00	0.02590
1.00	525.01	294.70	2.20	132.00	0.02514
1.25	656.26	294.20	2.70	162.00	0.02469
1.50	787.51	293.50	3.40	204.00	0.02590
1.75	918.76	292.80	4.10	246.00	0.02678
2.00	1050.02	292.30	4.60	276.00	0.02629
2.25	1181.27	292.00	4.90	294.00	0.02489
2.50	1312.52	291.40	5.50	330.00	0.02514
2.75	1443.77	290.50	6.40	384.00	0.02660
3.00	1575.02	290.00	6.90	414.00	0.02629
3.25	1706.28	289.40	7.50	450.00	0.02637
3.50	1837.53	288.90	8.00	480.00	0.02612
3.75	1968.78	288.60	8.30	498.00	0.02529

Table 1: Data for the experiment

From the above data, we get the mean value of the Verdet's constant to be $V_{\text{avg.}} = 2.571 \times 10^{-2}$ min/Oe/cm.



From the plot of $\Delta\theta$ and H, we obtain a linear fit with slope approx 0.2598 and we know that this slope corresponds to $V \times D$ where $D = 10$ cm for us. Hence from the fitting curve also, we get $V \approx 0.02598$.

4.2 Verification of Malu's Law

In the second part we fixed the polarisers, but varied the current and measured the change in intensity produced. The data is tabulated below.

Current (A)	Intensity (μA)
0.00	0.00
0.25	0.20
0.50	1.00
0.75	2.40
1.00	4.30
1.25	6.80
1.50	10.00
1.75	13.40
2.00	17.70
2.25	22.40
2.50	27.40
2.75	33.30
3.00	39.60
3.25	46.50
3.50	53.60
3.75	61.20

Table 2: Data Table for Malu's law

We plotted the measured intensity vs. the current on log-log scale and fitted with a linear function. The slope obtained is, 2.032 ± 0.008, which is in agreement with the expected value of 2 using Malu's law.

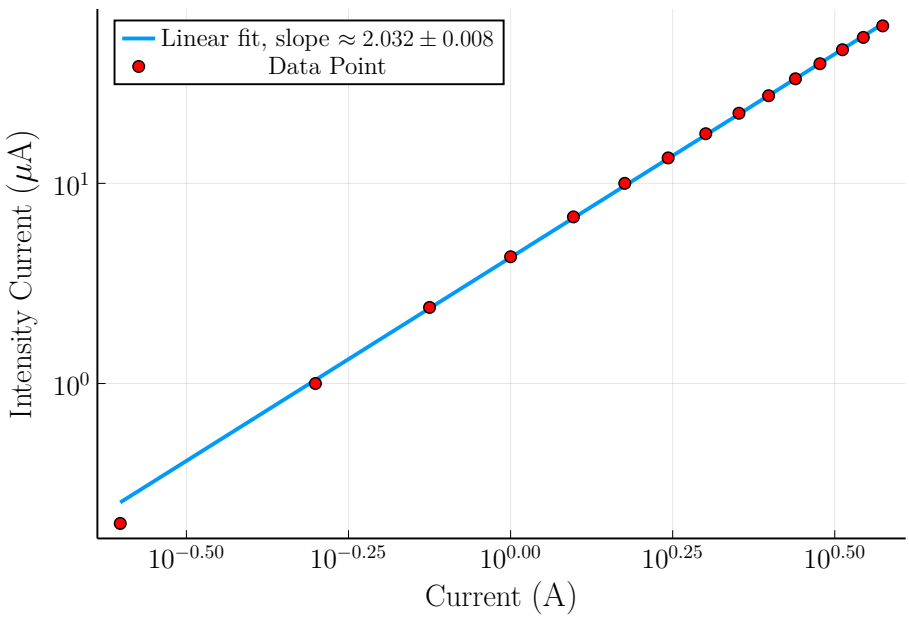


Figure 3: Intensity vs. Current in log-log scale with the linear fit shown with solid line.

5 Error Analysis

5.1 Random Error

From the formula of the Verdet constant, we get:

$$\frac{\Delta V}{V} = \frac{\Delta\theta}{\theta} + \frac{\Delta B}{B} = \frac{\Delta\theta}{\theta} + \frac{\Delta I}{I} \tag{9}$$

The least count of the current is 0.1 A, while that of the angle measurement in the polariser is $0.1^\circ \equiv 6'$. We take the first data point $I = 0.25$ A and $\theta = 30'$ and $V = 0.02286$ we get

$$\Delta V = 0.02286\left(\frac{6}{30} + \frac{0.001}{0.25}\right) = 0.0055$$

Hence, we can write the Verdet constant as $V = (0.02571 \pm 0.0055)$ min/Oe/cm. We can also calculate the accuracy of the experiment on Malu's law. The deviation in the experimentally derived slope from the theoretical value is,

$$\text{Error} = \frac{|2 - 2.032|}{2} \times 100\ \% \approx 1.6\%$$

5.2 Sources of Error

Sources of error for this experiment can be:

1. **Non-uniform Magnetic Field:** The experiment assumes that the magnetic field is uniform throughout the material. In reality, it is not perfectly uniform, which introduces error in the measurements.
2. **Non-monochromatic Light:** If the light is not monochromatic, different frequency components experience different rotations (due to different responses of RCP and LCP modes). Hence, the incident light should ideally be monochromatic.
3. **Thermal Fluctuation:** The Verdet constant and material properties can be temperature dependent. Thermal fluctuations can therefore lead to experimental errors.
4. **Alignment Issue:** If the laser is not properly aligned to pass horizontally through the medium, some intensity may be lost. This can lead to incorrect readings in the photodetector.
5. **Imperfect Polariser:** If the polarisers are not ideal, the light may not be perfectly polarised. This introduces noise and affects the accuracy of angle measurements.

6 Discussion and Conclusion

In this experiment, we investigated the Faraday effect in flint glass and verified Malus' law through intensity measurements. It was observed that the angle of rotation of the plane of polarisation increases linearly with the applied magnetic field, as expected theoretically. The Verdet constant was experimentally determined to be

$$V = (0.0257 \pm 0.0055) \text{ min/Oe/cm},$$

which is consistent across the measured range of currents.

In the second part, the variation of transmitted intensity with current was studied. The log-log analysis showed a power-law dependence with exponent 2.032 ± 0.008, confirming the quadratic relationship predicted by Malus' law under small-angle approximation.

Thus, the experiment successfully validates both the linear dependence of Faraday rotation on magnetic field and the quadratic dependence of transmitted intensity on the rotation angle.